



Saleh Eddine Khalfaoui

Software Engineer | Backend & Systems Engineering | AI Applications

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Profile

Software engineer based in Tunisia, focused on backend engineering, system design, and distributed architecture. Builds Java and Spring Boot microservices with Apache Kafka, REST APIs, and Retrieval-Augmented Generation (RAG). Also works across full-stack development, databases, and Docker. Final-year student at ESPRIT, available for a 2027 end-of-studies internship (PFE).

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, PHP, SQL

Backend: Spring Boot, NestJS, Node.js, Django, Laravel, Spring Cloud, REST APIs

Architecture: Microservices, Apache Kafka, event-driven architecture, distributed systems, system design, MVC

AI: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), AI-powered applications, OpenAI API

Frontend: React, Next.js, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB

DevOps: Docker, Git, GitHub, Linux, Jira, Agile/Scrum

Data and BI: ETL, Power BI

Professional Experience

Amen Bank | Software Engineering Intern

Tunis, Tunisia | June 2026 – August 2026

- Developed **Java/Spring Boot microservices** and **REST APIs** for an AI-powered banking application.
- Integrated **Apache Kafka** for asynchronous, event-driven communication between distributed services.
- Integrated **Retrieval-Augmented Generation (RAG)** and large language models into that backend.
- Containerized the services with **Docker** and kept boundaries decoupled and maintainable.

Technologies: Java, Spring Boot, Microservices, Apache Kafka, REST APIs, RAG, LLMs, Docker, Git

Mobelite Labs | Web & BI Developer

Tunis, Tunisia | February 2024 – January 2025

- Developed responsive interfaces in **React** and **Next.js**, integrated with **REST APIs**.
- Integrated **5 Power BI dashboards** with those APIs across **10K+ records**.
- Automated **ETL** workflows and reduced manual reporting time by **50%**.
- Improved decision-making efficiency by **30%** through centralized reporting, in Agile sprints with **Jira**.

Technologies: React, Next.js, REST APIs, Power BI, ETL, Jira

Amen Bank | Software Engineering Intern

Tunis, Tunisia | Summer 2025

- Developed web and backend features, including **REST APIs** and database-backed application logic.
- Implemented features from requirements and tested them for reliability and code quality in a banking environment.

Technologies: REST APIs, databases, testing

Selected Software Projects

ByteBattle — Competitive programming platform with AI-assisted features.

- Built challenge workflows, submission scoring, and leaderboards with **React**, **NestJS**, and **MongoDB**.
- Added **JWT** authentication, role-based access control, and **OpenAI API** integration in the NestJS service layer.
- Containerized the services with **Docker** for reproducible, deployment-ready environments.

Technologies: React, NestJS, MongoDB, Docker, REST APIs, JWT, OpenAI API

E-commerce microservices — Separate Spring Boot services with independent databases.

- Designed **Spring Boot** microservices with separate **PostgreSQL** databases and REST APIs.
- Implemented a **Spring Cloud** API gateway and containerized the stack with **Docker**.

Technologies: Spring Boot, Spring Cloud, PostgreSQL, Docker, REST APIs, Microservices

NutriTrace — Monolithic Laravel MVC application for traceability.

- Developed a **Laravel 12** MVC application in PHP for traceability, with relational **MySQL** models and **Leaflet** maps.
- Implemented authentication and **role-based access control** with **Spatie Permission** on a REST-oriented monolith.

Technologies: Laravel 12, PHP, MVC, MySQL, Spatie Permission, Leaflet, REST

CargoLink — Intelligent freight platform with map-based routing and RAG.

- Built a **Django** MVT application in Python with templates, the ORM, and HTML/CSS/JavaScript.
- Integrated **Leaflet** and the **OpenRouteService** API for route, distance, and ETA.
- Integrated a **Retrieval-Augmented Generation (RAG)** workflow for context-aware assistance grounded in logistics data.

Technologies: Django, Python, RAG, LLMs, Leaflet, OpenRouteService, HTML, CSS, JavaScript

Education

Engineering Cycle (Web & Software Engineering)

ESPRIT

Tunis | 2024–Present

Bachelor's Degree in Software Engineering and Information Systems

École Polytechnique Internationale Privée de Tunis

Tunis | 2021–2024

Languages

English — Fluent French — Fluent Arabic — Native.